Heart Attack Risk Prediction

Project overview

This proposal investigates whether demographic, clinical and lifestyle characteristics can be used to classify the binary Heart Attack Risk label (0 or 1) in the Heart Attack Risk Prediction Dataset. This is an educational analysis of synthetic records, not a diagnostic or clinical-risk calculator.

Dataset and source

The dataset is published on Kaggle by iamsouravbanerjee. Prior published analyses describe 8,763 rows and 26 columns; the actual file shape, values and quality will be verified directly once the original CSV is downloaded. Source: https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset

Research question and approach

Which recorded variables are associated with the dataset's heart-attack-risk label, and how accurately can supervised classifiers predict that label on previously unseen records? The notebook will separate 70% training and 30% testing before exploratory analysis. All descriptive plots, feature selection decisions and preprocessing estimates will use training data only.

Planned preparation and deliverables

Python will inspect duplicates, data types, missingness and target balance; parse systolic/diastolic blood pressure; generate simple engineered features; impute missing values; encode categorical fields; and scale numeric values. Matplotlib will compare training distributions before and after processing. Final outputs will include a reproducible notebook and cleaned training dataset.

Finals extension and evaluation

Finally, compare suitable classification models (such as logistic regression, decision tree and random forest) with the preserved test data, reporting confusion matrix, precision, recall, F1, ROC-AUC where appropriate and class balance. Clinical validity and external generalizability cannot be assumed from synthetic records.